Scope before Reuse: corrected theory and measured evaluation (draft for review)

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Abstract

Corrected problem formulation, theory and related-work paragraph for FSE-2027-13; see docs/paper/THEORY.md for the derivation and docs/paper/latex/CHANGES.md for what each part replaces.

1 Problem Formulation

Skills, scopes and cells. A skill is $S = \langle \pi, \mathcal{D}, \mathcal{I}, \mathcal{P}, \mathcal{K} \rangle$: a procedure π (a parameterised edit program), dependency and environment fingerprints $\mathcal{D}$, input assumptions $\mathcal{I}$ (executable predicates on the target), observed postconditions $\mathcal{P}$ with the contexts they were observed in, and an interaction footprint $\mathcal{K}$, the code regions and behavioural properties that π reads or writes. A context x is a (repository, revision, environment, task) tuple. A scope family $\mathbb{S} = \{s_1, \ldots, s_L\}$ is a finite partition of contexts declared before any trial. We call a pair $c = (S, s)$ a *cell*; for a set of skills whose footprints form a connected overlap graph, their composition, applied together in scope s, is a cell of its own (for two skills, $(S \circ S', s)$).

Loss and selective transfer risk. For a task x, $\ell(x, \pi) \in [0, 1]$ is its functional loss with the skill applied and $\ell(x, \varnothing)$ the loss without it. We use one minus the held-out pass rate plus 0.5 per regression, *clipped at* 1, so that the range assumption below holds. The paired difference is $Z = \ell(x, \pi) - \ell(x, \varnothing) \in [-1, 1]$.

Definition 1.1 (Selective transfer risk). For a cell $c = (S, s)$ and a distribution D over the contexts of scope s, $\mathrm{STR}_D(c) = \mathbb{E}_{x \sim D}\big[\ell(x, \pi) - \ell(x, \varnothing)\big]$. With P_s the distribution trials are drawn from and Q_s the deployment distribution, a cell is *admissible* if $\mathrm{STR}_P(c) \le -\theta$ for a benefit margin $\theta > 0$, and *harmful* if $\mathrm{STR}_Q(c) > 0$. After a retirement (Section 2) the environment may have changed; epoch k of a cell then has its own $P_s^{(k)}, Q_s^{(k)}$, and every statement below is per epoch.

The definition is an excess over the no-skill baseline, so a skill that neither helps nor harms is not admissible, and it is relative to a declared scope; Proposition 2.4 shows why it cannot be otherwise.

Assumptions.

Assumption 1 (Paired trials). *For every cell c there is an infinite sequence of tasks $x_1^c, x_2^c, \ldots$ drawn i.i.d. from P_s (in practice, drawn* with replacement *from the target's task set; without replacement the conditional mean of the next draw drifts and Lemma 2.1 needs a finite-population e-process instead); each task is run with and without the skill on the same snapshot and seed. Different cells may depend on each other.*

Assumption 2 (Bounded shift). *Deployment contexts of scope s follow Q_s with $\mathrm{TV}(P_s, Q_s) \le \xi_s$, where $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$.*

Execution rule. Skills that are executed together form a set E. Split E into the connected components of its footprint-overlap graph (two skills are adjacent when their footprints intersect). We execute E only if every component with two or more skills is itself an admitted cell; a skill whose composite with an already admitted skill is still undecided is not executed together with it. This closes the gap between per-cell certificates and what actually runs.

Admission problem. Given a library, a validation budget and a pre-allocated error budget α, decide which cells may execute, such that $\Pr[\exists \text{ an admitted cell } c : \mathrm{STR}_Q(c) > 0] \le \alpha$ over the whole library, over adaptively chosen cells and over re-admissions after retirement, and such that, under the execution rule, no executed set is harmful except on that event. Theorem 2.2 and Corollary 2.6 give this whenever $2\xi_s \le \theta$ for every scope; for coarser scopes they give the weaker, explicit statement $\mathrm{STR}_Q(c) \le -\theta + 2\xi_s$.

2 Theory

We claim no new statistics: the guarantee is assembled from betting e-processes [9], Ville's inequality [6] and a union bound. What is new is the estimand, the per-cell ledger that covers compositions and re-admission, the proof that the shift term is unavoidable, and the exact characterisation of composition risk.

2.1 Per-cell evidence

Lemma 2.1 (Betting e-process). *Let $Y_1, Y_2, \ldots$ take values in $[y_{\min}, y_{\max}]$ with $y_{\min} < \theta$, and suppose $\mathbb{E}[Y_i \mid \mathcal{F}_{i-1}] \le \theta$ for all i (the null H_0). The process $K_0 = 1$, $K_n = \prod_{i \le n}\big(1 + \lambda_i(Y_i - \theta)\big)$ with predictable $\lambda_i \in [0, 1/(\theta - y_{\min}))$ satisfies $\Pr[\exists n : K_n \ge 1/\beta] \le \beta$ for every $\beta \in (0, 1)$. For admission, $Y_i = -Z_i \in [-1, 1]$ and H_0^c is $\mathbb{E}[Y] \le \theta$ (the cell is not admissible), so $\lambda_i < 1/(1 + \theta)$.*

Proof. Since $Y_i \ge y_{\min}$, every factor is at least $1 - \lambda_i(\theta - y_{\min}) > 0$. By the null and predictability, $\mathbb{E}[K_i \mid \mathcal{F}_{i-1}] = K_{i-1}\big(1 + \lambda_i(\mathbb{E}[Y_i] - \theta)\big) \le K_{i-1}$, so (K_n) is a nonnegative supermartingale with $K_0 = 1$, and Ville's inequality gives the claim. □

The bound holds at any data-dependent stopping time, including the end of the validation budget, which is why a fixed-sample interval would not do. Under Assumption 1 the trials are i.i.d., so the null on the conditional mean reduces to $\mathbb{E}[Y] \le \theta$.

Remark 1 (Harm certificate used in the experiments). The experiments of Section 3 also use a certificate against *harm*: with $H_i \in \{0, 1\}$ indicating a harmful draw and a tolerated harm rate η, the null is $\Pr[H = 1] \ge \eta$. Setting $Y_i = 1 - H_i \in [0, 1]$ and $\theta = 1 - \eta$ gives factors $1 + \lambda(\eta - H_i)$, that is, $1 + \lambda\eta$ for a clean draw and $1 - c$ for a harmful one when $\lambda = c/(1 - \eta)$; the condition $\lambda < 1/(\theta - y_{\min}) = 1/(1 - \eta)$ is $c < 1$. Lemma 2.1 and Theorem 2.2 therefore apply unchanged, with "harm rate at least η" in place of "not admissible". With $c = 0.9$ and $\eta = 5\%$ the wealth grows only when the true harm rate is below $\log(1 + \lambda\eta)/(\log(1 + \lambda\eta) - \log(1 - c)) \approx 0.39\eta$, and certifying zero harm in a family of 16 cells at $\alpha = 0.1$ takes $\lceil \log(160)/\log(1 + \lambda\eta) \rceil = 110$ clean draws.

2.2 Library-wide admission guarantee

Theorem 2.2 (Admission over a declared scope family). *Let C be the countable set of all cells that could ever be formed, compositions included, and let a ledger $\alpha_{c,k} \geq 0$ be fixed in advance, as a function of the cell's identity, for every cell c and epoch $k \geq 0$ (epoch $k \geq 1$ is the k-th re-admission after retirement), with $\sum_{c \in C} \sum_{k \geq 0} \alpha_{c,k} \leq \alpha$; for instance $\alpha_{c,k} = \alpha_c \cdot 6/(\pi^2(k+1)^2)$ with $\sum_c \alpha_c \leq \alpha$. A library that grows is covered as long as the share of every future cell is fixed by the same rule. Suppose each epoch uses fresh tasks and admits c only when its e-process (Lemma 2.1) reaches $1/\alpha_{c,k}$, while the choice of cells, their order and the stopping times may depend on all data. Under Assumptions 1 and 2,*

$$\Pr\big[\exists \text{ admitted } (c,k) : \ \mathrm{STR}_{Q^{(k)}}(c) > -\theta + 2\xi_{s(c)}\big] \leq \alpha,$$

where $P^{(k)}, Q^{(k)}$ are the trial and deployment distributions of epoch k. In particular, if $2\xi_s \leq \theta$ for every scope, the probability of admitting any harmful cell is at most α.

Proof. For every (c,k) let $A_{c,k} = \{\sup_n K_n^{c,k} \geq 1/\alpha_{c,k}\}$, defined on the whole infinite task sequence of that epoch. An admission of (c,k) implies $A_{c,k}$; adaptive selection and stopping can only forgo admissions. Hence the event that some cell with H_0^c true is admitted is contained in $\bigcup_{(c,k):H_0^c} A_{c,k}$, whose probability is at most $\sum \alpha_{c,k} \leq \alpha$ by Lemma 2.1 and a union bound, which needs no independence across cells. Finally, by Lemma 2.3, $\mathrm{STR}_{Q^{(k)}}(c) > -\theta + 2\xi_s$ implies $\mathrm{STR}_{P^{(k)}}(c) > -\theta$, that is, H_0^c for that epoch. If $2\xi_s \leq \theta$, then $\mathrm{STR}_Q(c) > 0$ implies $\mathrm{STR}_Q(c) > -\theta + 2\xi_s$. □

Remark 2 (Data-dependent budgets). The ledger above is fixed in advance, which is what makes the union bound free of any independence assumption. A budget that depends on data (for example, α_c proportional to an estimated reuse frequency) remains valid if (i) the total is at most α almost surely and (ii) the trial sequence of cell c is independent of the information used to set α_c: then $\Pr[A_{c,k} \mid \alpha_{c,k}] \leq \alpha_{c,k}$ and summing the expectations gives α. Without (ii), for instance when two cells share tasks, a data-dependent budget can break the guarantee.

Lemma 2.3 (Shift). *If f takes values in an interval of length b, then $|\mathbb{E}_Q f - \mathbb{E}_P f| \leq b \cdot \mathrm{TV}(P,Q)$. Since $Z \in [-1,1]$, $|\mathrm{STR}_Q(c) - \mathrm{STR}_P(c)| \leq 2\,\mathrm{TV}(P_s, Q_s) \leq 2\xi_s$, and the constant 2 is attained.*

Proof. Centre f so that $|g| \leq b/2$; then $|\mathbb{E}_Q g - \mathbb{E}_P g| \leq (b/2)\sum_x |Q(x) - P(x)| = b\,\mathrm{TV}(P,Q)$. For attainment, take task types u with $Z = 1$ and d with $Z = -1$, $P(u) = 0.1$, $Q(u) = 0.2$: then TV = 0.1 and the risks differ by 0.2. □

Why the budget is per cell. Allocating one α_s per *scope* and letting every skill in the scope use the threshold $1/\alpha_s$ does not control the family-wise error: with ten skills at the null boundary, $\theta = \alpha = 0.1$ and 300 trials, the family-wise false admission rate of our e-process is 0.45, against 0.05 with the per-cell budget $\alpha/10$ (Section 2.6).

2.3 Scope and shift are necessary

Assumption 3 (Rich target class). *For every value of the features a procedure observes (fingerprints, predicates, trial outcomes on the observed repositories), there is a repository with those features whose task losses are arbitrary in $[0,1]$.*

Proposition 2.4 (No guarantee without target evidence or bounded shift). (a) *Let a procedure decide from data of observed repositories $\mathcal{R}_{\mathrm{obs}}$ only, and admit a skill S with positive probability. Under Assumption 3 there is a repository $R^\dagger \notin \mathcal{R}_{\mathrm{obs}}$ with $\mathrm{STR}(S, \{R^\dagger\}) = 1$ on which the admission probability is unchanged.* (b) *For every $\theta \in (0,1)$, $\delta > 0$ with $\theta + \delta < 1$, $\xi \in (0, 1/2]$ and every procedure, there are P, Q with $\mathrm{TV}(P,Q) = \xi$ and a cell c with $\mathrm{STR}_P(c) = -\theta - \delta$ and $\mathrm{STR}_Q(c) = \mathrm{STR}_P(c) + 2\xi$, whose admission probability does not depend on Q.*

Proof. (a) By Assumption 3, take $R^\dagger$ that has the same values of every feature the procedure reads (the data of $\mathcal{R}_{\mathrm{obs}}$ and the target's own fingerprints and predicates) and has $\ell(x,\pi) = 1$, $\ell(x,\varnothing) = 0$ for every task. The procedure's input, hence its admission probability, is unchanged, and $Z \equiv 1$ on $R^\dagger$. (b) Take task types u ($Z = 1$) and d ($Z = -1$) with $P(d) = (1 + \theta + \delta)/2 \geq \xi$, and obtain Q by moving mass ξ from d to u. The procedure only reads tasks drawn from P. □

Part (b) shows that no procedure that learns only from P can certify Q-risk better than $\mathrm{STR}_P + 2\xi$, so the shift term of Theorem 2.2 cannot be removed or reduced; this is why the guarantee is relative to a declared scope family and why coarse scopes make it weak.

2.4 Composition

Theorem 2.5 (Composition risk is not controlled by individual risks). *For $a, b \in [-1, 0)$,*

$$\sup\big\{\mathrm{STR}(S_1 \circ S_2) : \ \mathrm{STR}(S_1) = a, \ \mathrm{STR}(S_2) = b\big\} = 1 + \min(a,b),$$

and the supremum is attained. In particular, two skills that each meet the margin $-\theta$ can compose to excess risk $1 - \theta > 0$, and no bound on the composite risk improves on the trivial one implied by $\ell \in [0,1]$.

Proof. Upper bound, for any skills: $\mathrm{STR}(S_1 \circ S_2) \leq 1 - \mathbb{E}\ell(\varnothing)$, and $\mathbb{E}\ell(\varnothing) \geq -a$ because $\mathbb{E}\ell(\pi_1) = \mathbb{E}\ell(\varnothing) + a \geq 0$; likewise $\geq -b$. Attainment, with losses realizable by Assumption 3: constant losses $\ell(\varnothing) = -\min(a,b)$, $\ell(\pi_1) = \ell(\varnothing) + a$, $\ell(\pi_2) = \ell(\varnothing) + b$, $\ell(\pi_1 \circ \pi_2) = 1$. □

Theorem 2.5 says nothing about footprints. Screening only pairs whose footprints overlap is therefore sound only under an additional assumption, which we state and test rather than presume.

Assumption 4 (Non-interaction of disjoint footprints). *For sets of skills A, B whose footprints are disjoint ($\mathcal{K}_A \cap \mathcal{K}_B = \varnothing$, with $\mathcal{K}_A$ the union over A), and every task x, $\ell(x, \pi_A \circ \pi_B) - \ell(x,\varnothing) = [\ell(x,\pi_A) - \ell(x,\varnothing)] + [\ell(x,\pi_B) - \ell(x,\varnothing)]$.*

Corollary 2.6 (What actually runs). *Under Assumption 4 and the execution rule of Section 1, outside the event of Theorem 2.2 (probability at most α), every executed set E with m footprint-connected components satisfies $\mathrm{STR}_Q(E) \leq m(-\theta + 2\xi_s)$; in particular no executed set is harmful when $2\xi_s \leq \theta$. Only footprint-overlapping components need a composite cell: with mean overlap degree $\bar{d}$ there are $|\mathcal{L}_s|\,\bar{d}/2$ overlapping pairs per scope.*

Proof. Components have pairwise disjoint footprints, so applying Assumption 4 $m - 1$ times and taking expectations under Q gives $\mathrm{STR}_Q(E) = \sum_{j=1}^m \mathrm{STR}_Q(E_j)$. Each E_j is an admitted cell (a

single skill, or a composite by the execution rule), so outside the event of Theorem 2.2 each term is at most $-\theta + 2\xi_S$. □

Assumption 4 is strong: two edits in different functions can still interact through shared state that no footprint records. We therefore report it as an assumption and measure how often it fails on pairs with disjoint footprints, rather than presume it.

2.5 The value of pairing

Proposition 2.7 (Pairing). *Let* $V_\pi = \mathrm{Var}\,\ell(x,\pi)$, $V_\varnothing = \mathrm{Var}\,\ell(x,\varnothing)$ *and* $C = \mathrm{Cov}(\ell(x,\pi),\ell(x,\varnothing))$ *on the same task. Estimating* STR *from* n *paired tasks has variance* $(V_\pi + V_\varnothing - 2C)/n$*; from two independent arms of* n *tasks each it has variance* $(V_\pi + V_\varnothing)/n$*. Both designs cost* $2n$ *runs, so pairing reduces the number of tasks needed for a given precision by* $R = (V_\pi + V_\varnothing)/(V_\pi + V_\varnothing - 2C)$*. Under the additive model* $\ell(\varnothing) = T$, $\ell(\pi) = T + S$ *with independent* T, S, $R = 1 + 2\sigma_T^2/\sigma_S^2$.

Proof. Direct computation of the two variances; $V_\pi, V_\varnothing$ include any run-to-run noise, and only the shared task component enters C. Under the additive model $V_\pi + V_\varnothing = 2\sigma_T^2 + \sigma_S^2$ and $C = \sigma_T^2$. □

2.6 Numerical checks

Every statement above is checked numerically by a script in the artifact, using the e-process of our implementation: per-cell validity, the failure of a per-scope budget, the attained constant 2 in Lemma 2.3, the bound of Theorem 2.5 on random instances, the pairing ratio, the convergence of the ledger, and the failure of re-admission without a ledger (re-testing a retired cell with the full α at every epoch).

3 Evaluation

All studies were pre-registered: protocol, configuration and tools were hashed before any outcome was seen, seeds were drawn from the public drand beacon after the freeze, and amendments are recorded and labelled. Results that were analysed after seeing data are marked as such. Every number below is produced by a workflow in the artifact.

3.1 Where simple gates fail: the phase diagram

For a patch that always helps, Table 1 gives the probability that each gate admits it as a function of the true harm rate h ($\eta = 5\%$, $\alpha = 0.1$; exact for the smoke, zero-tolerance and fixed-sample gates, Monte Carlo for the sequential rules). A zero-tolerance gate with 300 draws keeps fewer than half of the acceptable patches once $h > 0.23\%$; a 20-draw smoke test still admits more than 10% of inadmissible patches until $h \approx 11\%$. Outside the band $[3/n,\ 1 - 0.1^{1/20}] \approx [0.75\%, 10.9\%]$ ($n = 400$) a simple gate suffices. At the same number of draws a fixed-sample test is as safe as the sequential certificate and *more* powerful (0.92 versus 0.65 at $h = 2\%$, 300 draws). What the sequential rule buys is validity at data-dependent stopping and early stopping (110 clean draws in a 16-cell family), not power. On synthetic targets with rare harm (Study H, 50 repetitions per cell, 300 draws), the certificate admitted 0/100 cells with $\rho \geq \eta$, while a 5-draw smoke test admitted 38/50 and 27/50 at $\rho = 5\%$ and 10%, and replay plus fingerprints admitted every cell.

Table 1: Probability that a gate admits an always-helpful patch, by true harm rate h ($h \geq \eta$: admitting is an error).

h	smoke 20	zero-tol. 300	fixed 300	certificate, 300	mixture, 1000
0.3%	0.94	0.41	1.00	1.00	1.00
1%	0.82	0.05	1.00	0.95	1.00
2%	0.67	0.00	0.92	0.65	1.00
3%	0.54	0.00	0.59	0.36	0.78
5% (error)	0.36	0.00	0.06	0.10	0.10
10% (error)	0.12	0.00	0.00	0.01	0.00

3.2 Skill reuse on real targets

Two skill families, both frozen AST rewrites (excluding `bool` from `int` checks, learned from one PySStuBs fix, and `== None`→`is None`), were applied to 14 real targets in three pre-registered studies (28 cells). *Study D* (8 targets, 16 cells): ground truth from 300 draws plus the target's full test suite gave 3 beneficial, 9 unchanged, 4 out-of-scope and 0 harmful cells. B2, replay plus fingerprints (B6-R) and always-retest (B8) each admitted 12 cells, the 3 beneficial ones and all 9 unchanged ones; the certificate (benefit e-process with e-Holm over the 16 cells, family-wise $\alpha = 0.1$) admitted 2 cells, both beneficial, after 110 draws each, and missed the third, whose mean benefit (−0.11) is smaller than the margin $\theta = 0.2$. It used 3,220 trial runs against 3,600 for B8. *Studies E and G*: on the four harmful cells (`pyyaml`, `packaging`, `databricks`: harm on 87–100% of inputs; `pydal`: 81/300), B2 and B6-R admitted all four and the certificate rejected all four, within two draws for the first three. The targets' own test suites (1,287, 51,524 and 130 tests) showed *no* new regression in Study E; in Study G they did (21 new failures). Re-running 2,400 recorded draws in shuffled order with fresh workers reproduced every paired outcome, so the draws behave as independent, as Assumption 1 requires. *Reading.* On these targets harm was frequent and deterministic, so a simple gate with target evidence suffices; the certificate's distinctive behaviour is refusing changes that are not beneficial, not catching rare harm.

3.3 Partial harm in LLM patches

Study P drew fresh repository changes from a pre-registered frame of 3,562 candidates; for each usable instance a light model generated one patch, and 205 patches were plausible (they pass the hidden tests). With a function-level oracle (400 inputs per patch), 16/205 = 7.8% (Wilson 4.9–12.3%) had an estimated regression rate inside the partial-harm band. The test-based census admitted all 16 patches with $\hat{\theta} > 5\%$, a 20-draw smoke test admitted 2, and the zero-tolerance, fixed-sample and sequential gates admitted none. A pre-registered manual check (RQ5) then found that 97.5% of the "regression" inputs were inputs the developers' fix also rejects; counting only regressions on inputs the fix accepts left 1 of 205 patches as planned, 3 when deliberate rejections that the patch skips are added (after seeing data), and at most 6 when non-reproducible patches are counted.

3.4 Naive differences versus contract regressions on SWE-bench Verified

Study Q tested the RQ5 finding on an independent, widely used benchmark. For 469 SWE-bench Verified instances with at least

one public resolved patch, up to three distinct resolved patches were scored in the official Docker images (245 instances usable, 694 patches). 43.1% of patches behaved differently from the developers' fix on some input; 11.1% (Wilson 9.0–13.6%) had a *contract regression*, an input on which the fix returns a value and the patch does not, or the fix rejects on purpose and the patch accepts. The confirmatory hypothesis, that most naive differences are not contract regressions, was *not* supported: 6/55 (10.9%, Wilson 5.1–21.8%; one-sided $p \approx 1$). A manual review added after the result (labelled as such) re-ran the 55 units with outputs kept: 36 reproduced exactly, and two independent raters agreed on 87.7% of 154 items. 8/36 (22.2%, Wilson 11.7–38.1%) units had no regression confirmed by both raters; excluding three units whose inputs are model-built stand-in objects gives 11/36 (18.0–46.9%). The RQ5 conclusion therefore does not generalise: on SWE-bench Verified most naive differences are real, observable behaviour differences from the fix. Whether they are *wrong* is a separate question that a reference implementation cannot answer.

3.5 Who defines harm

On 14 measurable targets of Studies D, E and G, a model-written input generator agreed with the authors' hand-written oracles on 11/14 detection decisions (Cohen's $\kappa = 0.57$), a signature-based generator on 8/14 ($\kappa = 0.19$). In one disagreement the generated inputs were right: on `cerberus` the skill rewrote two functions other than the intended one; the hand-written oracle, which checks the intended region, saw nothing, while generated inputs changed behaviour on 24–56 of 200 inputs. An oracle written from the author's intent misses side effects outside that intent, which is the scope problem this paper is about.

4 Threats to Validity

Assumptions of the guarantee. Theorem 2.2 assumes i.i.d. trials drawn with replacement (Assumption 1); we checked that the harness is stateless (2,400 re-run draws, 0 differences) but not that real task streams are identically distributed. The shift bound ξ_s (Assumption 2) is declared, not estimated; the guarantee excludes harmful admissions only when $2\xi_s \le \theta$, and Proposition 2.4 shows that no procedure learning from trials alone can do better. Screening only footprint-overlapping pairs relies on Assumption 4, which *we have not tested*: two edits in different functions can still interact through state no footprint records. The execution rule is a design constraint that a deployment must enforce.

Construct validity. "Harm" is a difference from a reference: a hand-written oracle (Studies D–H), or the developers' fix (Studies P, Q). A difference from the fix is not necessarily a defect: in the manual review some patches arguably behaved better than the fix. Classifying an input as valid because the fix returns a value overestimates contract regressions when generated inputs are stand-in objects (three of 31 reviewed units); the sensitivity analysis keeps the conclusion. Hand-written oracles check the author's intent and miss side effects outside it (`cerberus`).

Internal validity. The second rater shares a model family with the input generator, and sees outputs but not inputs. 19/55 re-run units did not reproduce because the model-written generator changed; they enter as bounds. The model-written generator sees the fix's diff, so inputs concentrate near the changed region. Four Study Q instances failed for a tool limitation (patches that create files) and count as unusable. The Study Q analysis script defined the inflation ratio differently from the protocol; we report both.

External validity. The skill studies cover two hand-written rewrite families on 14 targets; we make no claim about learned skill libraries at scale, compositions in the wild, or retirement beyond a constructed demonstration. In Study P half of the in-band cases come from repositories that look like student assignments; Study Q covers public resolved submissions to one benchmark.

5 Related work (excerpt)

5.1 Risk control, e-processes and multiple testing

Conformal prediction [7], conformal risk control [2] and Learn-Then-Test [1] give distribution-free guarantees under exchangeability. Skill transfer violates exchangeability across repositories, so our guarantee is relative to a declared scope family (Proposition 2.4). For sequential evidence we use betting e-processes [9] and Ville's inequality [6], which make an admission valid at any data-dependent stopping time [4, 5]. Simultaneous control over the library is family-wise (FWER) and comes from a union bound over cells; it requires no independence. A false-discovery-rate variant is available through e-BH [8], at the price of a weaker, FDR-type guarantee; Benjamini–Hochberg [3] is used only for the secondary comparisons of the evaluation.

References

[1] Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. 2021. Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control. arXiv:2110.01052.

[2] Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. 2024. Conformal Risk Control. In *International Conference on Learning Representations (ICLR)*.

[3] Yoav Benjamini and Yosef Hochberg. 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society Series B* 57, 1 (1995), 289–300.

[4] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. 2021. Time-Uniform, Nonparametric, Nonasymptotic Confidence Sequences. *The Annals of Statistics* 49, 2 (2021), 1055–1080.

[5] Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. 2023. Game-Theoretic Statistics and Safe Anytime-Valid Inference. *Statist. Sci.* 38, 4 (2023), 576–601.

[6] Jean Ville. 1939. *Étude critique de la notion de collectif.* Gauthier-Villars, Paris.

[7] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. 2005. *Algorithmic Learning in a Random World.* Springer.

[8] Ruodu Wang and Aaditya Ramdas. 2022. False Discovery Rate Control with E-values. *Journal of the Royal Statistical Society Series B* 84, 3 (2022), 822–852.

[9] Ian Waudby-Smith and Aaditya Ramdas. 2024. Estimating Means of Bounded Random Variables by Betting. *Journal of the Royal Statistical Society Series B* 86, 1 (2024), 1–27.